

Building a Campus Cloud with Clientless VM Access: Proxmox VE, Guacamole, RBAC, and Tailscale

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Agenda

01 Problems & Solutions

02 System Architecture

03 System Setup

04 Demonstration

05 Q&A

Problems & Solutions



Problems & Solutions

01 • ACCESS & ACCOUNTS

PROBLEM VPN required per student

SOLUTION

Direct Web Access (No VPN)

02 • SELF-MANAGEMENT

PROBLEM Only provided VMs

SOLUTION

Snapshot & Monitoring

03 • INTEGRATED SUPPORT

PROBLEM Confused to report issues

SOLUTION

Integrated Requests & Tickets

04 • INSTANT PROVISIONING

PROBLEM Slow manual VM creation
* Manual setup from ISO / Clone

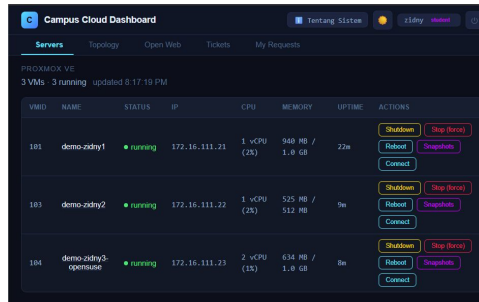
SOLUTION

Templates & Auto-Clone
* Linked-Clone support

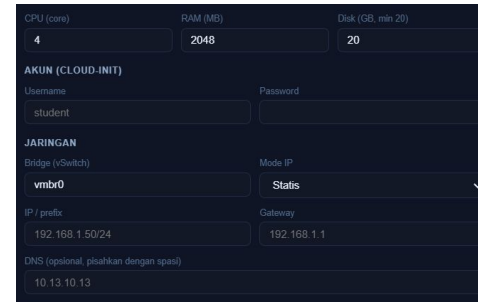
System Architecture

Dashboard Overview

Student Dashboard

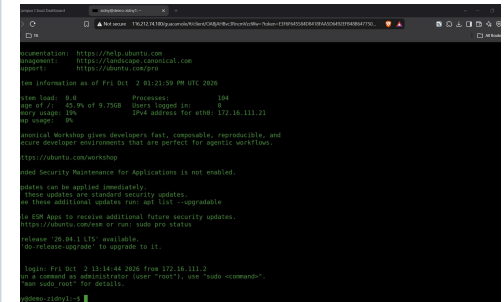


Create VM Form

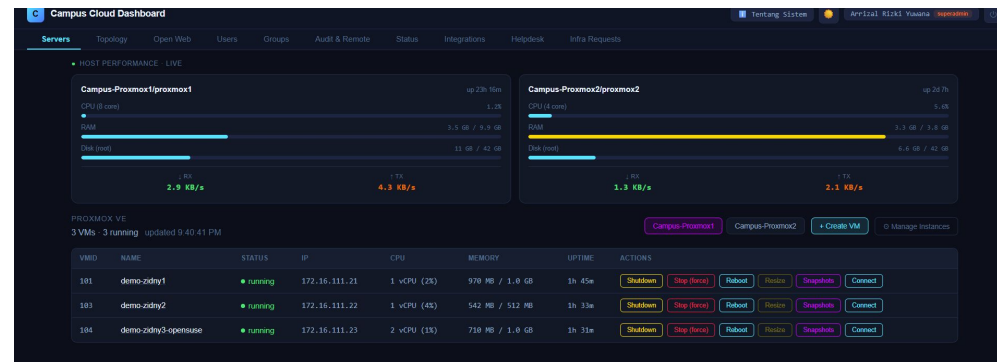


The Create VM Form includes fields for CPU (core), RAM (MB), and Disk (GB, min 20). It also has sections for AKUN (CLOUD-INIT) with Username and Password, and JARINGAN (Network) with Bridge (vSwitch), Mode IP, IP / prefix, Gateway, and DNS (optional).

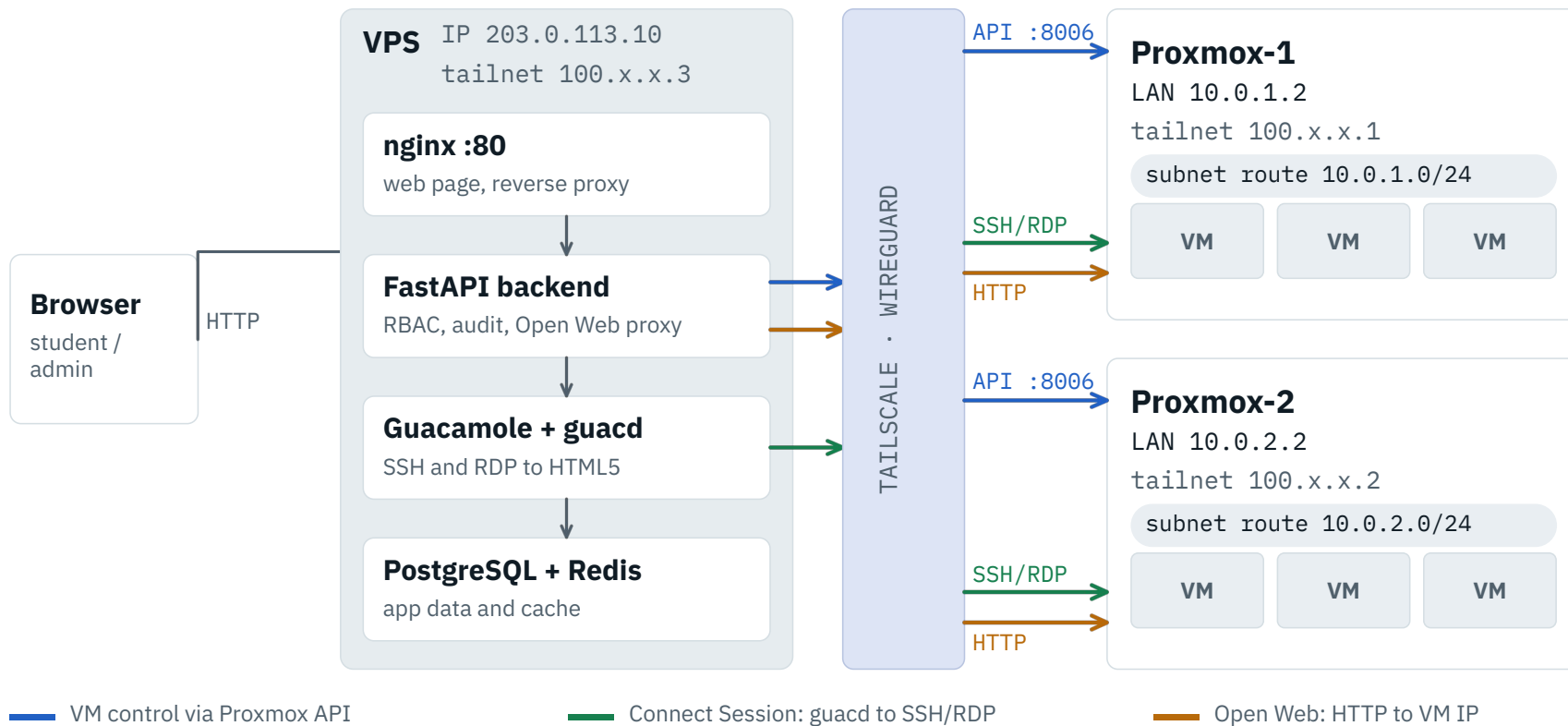
Clientless Access



Admin Dashboard



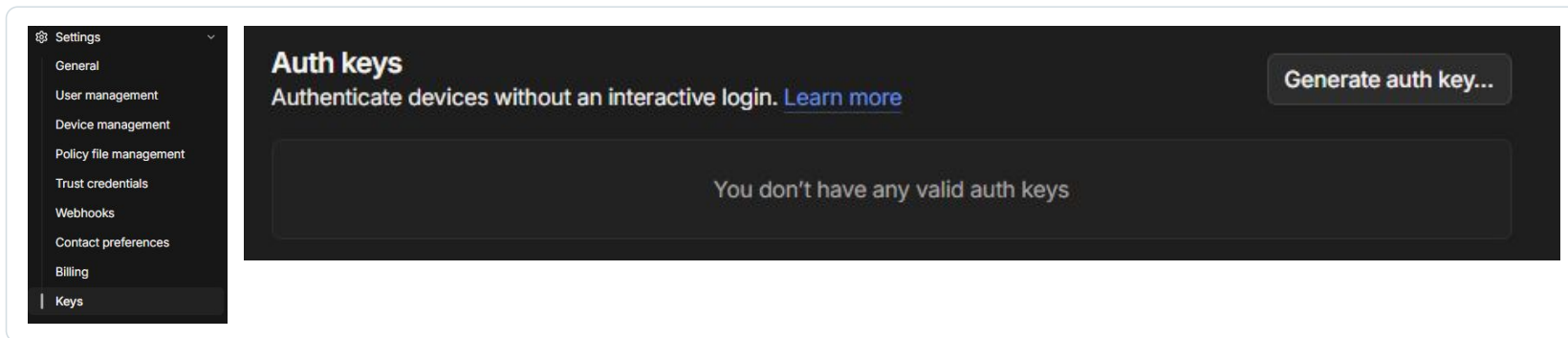
System Architecture



System Setup

Connecting VPS and Proxmox with Tailscale

TAILSCALE: GENERATE AUTH KEYS



The screenshot shows the Tailscale web interface. On the left is a dark sidebar with a 'Settings' menu. The 'Keys' option is selected and highlighted. The main content area has a dark background and is titled 'Auth keys'. Below the title is a link to 'Learn more' and a 'Generate auth key...' button. A large, empty rectangular box contains the text 'You don't have any valid auth keys'.

Settings

- General
- User management
- Device management
- Policy file management
- Trust credentials
- Webhooks
- Contact preferences
- Billing
- Keys

Auth keys

Authenticate devices without an interactive login. [Learn more](#)

Generate auth key...

You don't have any valid auth keys

⚠ Key only view one time

```
tskey-xxxxxx-xxxxxxxxxxxxxxxx-xxxxxxxxxxxxxxxx
```

Connecting VPS and Proxmox with Tailscale

- 1 VPS and each Proxmox** Install Tailscale and log in to the same tailnet.

```
curl -fsSL https://tailscale.com/install.sh | sh  
tailscale up --authkey=tskey-xxxxxx-xxxxxxxxxxxxxxxx-xxxxxxxxxxxxxxxx
```
- 2 Each Proxmox** Enable IP forwarding so Proxmox can forward traffic to VMs.

```
echo 'net.ipv4.ip_forward = 1' > /etc/sysctl.d/99-tailscale.conf  
sysctl -p /etc/sysctl.d/99-tailscale.conf
```
- 3 Each Proxmox** Advertise its VM subnet (example for Proxmox-1).

```
tailscale up --advertise-routes=10.0.1.0/24
```
- 4 Tailscale Admin Console** Approve route: Machines, select Proxmox node, Edit route settings, check its subnet.
- 5 VPS** Accept routes, then test: route to VM IP must go through tailscale0.

```
tailscale up --accept-routes  
ip route get <IP-VM>
```

Connecting VPS and Proxmox with Tailscale

Approve route: Machines, select Proxmox node, Edit route settings, check its subnet.

MACHINE	ADDRESSES ⓘ	VERSION	LAST SEEN	
proxmox1 s.*****r@gmail.com Subnets	100.81.216.103 ▾	1.102.4 Linux 7.0.2-6-pve	● Connected	Share... ⋮
proxmox2 s.*****r@gmail.com Subnets	100.76.245.120 ▾	1.102.4 Linux 7.0.2-6-pve	● Connected	⋮
campusclouddashboard s.*****er@gmail.com	100.73.245.115 ▾	1.102.4 Linux 6.8.0-111-generic	● Connected	⋮

Edit route settings of proxmox1

 **Key expiry is enabled**
If this machine's key expires, your relayed traffic may be interrupted until you reauthenticate.

Subnet routes
Connect to devices you can't install Tailscale on by advertising IP ranges as subnet routes. [Learn more](#)

☒ 10.0.1.0/24

Exit node
Allow your network to route internet traffic through this machine. [Learn more](#)

☐ Use as exit node ⓘ

[Cancel](#) [Save](#)

Route Approval Instructions

1. Locate Node

Go to Machines tab in Tailscale Admin Console and select your target Proxmox node.

2. Access Route Settings

Click the context menu (...) on the right side and choose "Edit route settings".

3. Approve Subnet

Check the box for the advertised subnet (e.g., 10.0.1.0/24) to enable traffic routing.

Installing Campus Cloud Dashboard on VPS

1. Prepare Linux VPS

Ubuntu, Debian, or openSUSE, with root or sudo access and port 80 available. Docker does not need to be installed beforehand.

2. Clone Repository & Install

```
git clone https://github.com/Arrizal2004/CCD campus-cloud-dashboard
cd campus-cloud-dashboard
./setup.sh
```

What setup.sh does

- Checks and installs Docker if not present
- Creates .env with random secrets and passwords
- Ensures port 80 is not in use
- Builds images one by one, then runs all containers

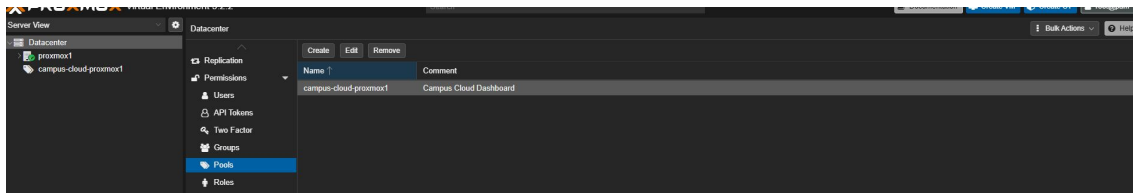
```
root@campusclouddashboard:~# docker ps
```

CONTAINER ID	IMAGE	COMMAND	CREATED	STATUS	PORTS	NAMES
ce24aa7ef7c2	campus-cloud-dashboard-frontend	"/docker-entrypoint...."	22 hours ago	Up 22 hours	0.0.0.0:80->80/tcp, [::]:80->80/tcp	ccd-frontend
432cfee808bd	campus-cloud-dashboard-backend	"uvicorn main:app --..."	22 hours ago	Up 22 hours	127.0.0.1:8001->8000/tcp	ccd-backend
e2a759bf454e	guacamole/guacamole:1.5.5	"/opt/guacamole/bin/..."	22 hours ago	Up 22 hours (healthy)	127.0.0.1:8081->8080/tcp	ccd-guacamole
8e0b70d8717f	postgres:16-alpine	"docker-entrypoint.s..."	22 hours ago	Up 22 hours (healthy)	127.0.0.1:5433->5432/tcp	ccd-postgres
a6ae332d7792	redis:7-alpine	"docker-entrypoint.s..."	22 hours ago	Up 22 hours (healthy)	127.0.0.1:6380->6379/tcp	ccd-redis
7edaa1afa872	guacamole/guacd:1.5.5	"/bin/sh -c '/opt/gu..."	22 hours ago	Up 22 hours (healthy)	127.0.0.1:4823->4822/tcp	ccd-guacd

On Proxmox: Create Pool and API Token

1. Create pool in Proxmox

```
pveum pool add campus-cloud --comment  
"Campus Cloud Dashboard"
```



2. Create API Token

```
pveum user token add root@pam  
ccd-dashboard --privsep 1
```

key	value
full-tokenid	root@pam!ccd-dashboard
info	{"privsep": "1"}
value	726fcee1-xxxx-xxxx-xxxx-7d27018d0ffd

--privsep 1 is important: the token does not automatically get root privileges and strictly obeys manually granted ACLs. Without privsep, access is equivalent to root—do not use it.

3. Configured in CCD

Token ID: root@pam!ccd-dashboard

Token Secret: 726fcee1-xxxx-xxxx-xxxx-7d27018d0ffd

On Proxmox: Permissions (ACL) for Token

REQUIRED

```
# Full control over campus-cloud pool (only within this pool)
pveum acl modify /pool/campus-cloud --tokens 'root@pam!ccd-dashboard' --roles PVEAdmin

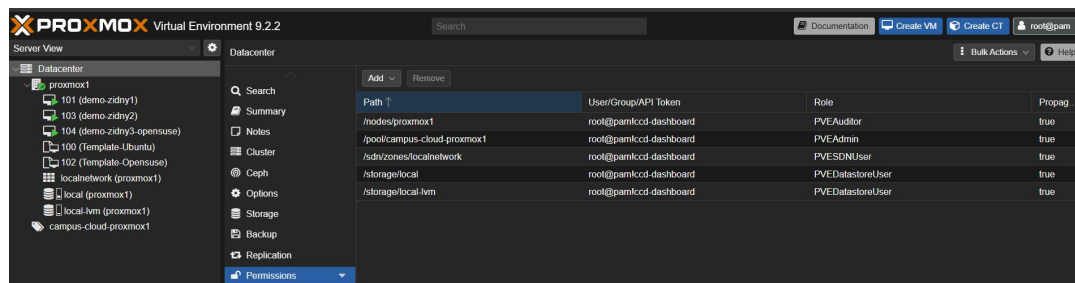
# Read and write VM disks in the storage where VM is stored
pveum acl modify /storage/<storage-disk> --tokens 'root@pam!ccd-dashboard' --roles PVEDatastoreUser
```

```
# SDN zone
pveum acl modify /sdn/zones/<zone-name> --tokens 'root@pam!ccd-dashboard' --roles PVESDNUser

# For Host Performance panel in dashboard
pveum acl modify /nodes/<node-name> --tokens 'root@pam!ccd-dashboard' --roles PVEAuditor
```

VERIFICATION

```
pveum acl list
```



Path	User/Group/API Token	Role	Propag...
/nodes/proxmox1	root@pam!ccd-dashboard	PVEAuditor	true
/pool/campus-cloud-proxmox1	root@pam!ccd-dashboard	PVEAdmin	true
/sdn/zones/localnetwork	root@pam!ccd-dashboard	PVESDNUser	true
/storage/local	root@pam!ccd-dashboard	PVEDatastoreUser	true
/storage/local-lvm	root@pam!ccd-dashboard	PVEDatastoreUser	true

Adding Proxmox to CCD

1. Open Integrations

Log in as admin, then open the Integrations tab.

2. Add Instance

Manage Instances, then + Add Instance, and fill form.

Form Details

Label	lab (any name)
Host	100.x.x.1:8006 (tailnet IP)
Token ID	root@pam!ccd-dashboard
Token Secret	from token creation step
Verify SSL	disable if self-signed certificate

CLOUD MANAGEMENT DEMO OPENSUSE SUMMIT ASIA 2026

Fri, 9 Oct 2026 12:39:15 GMT+7

AboutIDEN

Apprial Rizki YuhanaSuperadmin

ServersTopologyOpen WebUsersGroupsAudit & RemoteStatusIntegrationsSystemHelpdeskInfra Requests

Due to electrical maintenance, the server will be down from 06:00 to 15:00 (please back up your work).

PROXMOX INSTANCES

Campus - Proxmox1

100.81.216.103:8006 · root@pam/ccd-dashboard · SSL not verified

SSH terminalEditDelete

Campus - Proxmox2

100.78.245.120:8006 · root@pam/ccd-dashboard · SSL not verified

SSH terminalEditDelete

The SSH terminal to the Proxmox host opens in a new tab through Guacamole. The username and password are asked for there every time and are not stored in the dashboard. Only sysadmins and superadmins can open it.

+ Add instance

Role Based Access Control

Feature / Capability	Admin (superadmin, sysadmin)	Student
View VM list	All VMs	Only owned VMs
Start, stop, snapshot, rollback	All VMs	Owned VMs (Delete snapshot: admin only)
Create, delete, resize VM	Yes	No
Manage users, groups, assignments	Yes	No
Proxmox host list & performance	Yes	No
Audit logs, kill sessions, revoke links	Yes	No
Tickets and VM requests	Manage	Create

Hierarchy Note: superadmin > sysadmin > student. Unverified students can only access their request page.

(Demonstration) Cloud management openSUSE Summit Asia 2026

<https://youtu.be/wkvkfp-e-4>

Github Documentation

<https://github.com/arrizal2004/ccd>

LinkedIn

<https://www.linkedin.com/in/arrizalrizkiyuwana/>

Thank You



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